Network Traffic Analysis & Threat Detection Lab

Objective

To build a small virtual network using a Windows Host and Kali Linux VM and perform packet analysis, reconnaissance detection, and service enumeration using Wireshark and Nmap.

Lab Environment

- Windows 11 Host
- Kali Linux VM
- VirtualBox Host-Only Network
- Wireshark
- Nmap

Network Topology

Windows Host (192.168.56.1)

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Host-Only Network

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Kali Linux VM (192.168.56.102)

Activities Performed

1. ARP Analysis

Observed ARP requests and replies used for MAC address resolution.

2. ICMP Analysis

Captured Echo Request and Echo Reply packets to verify connectivity.

3. TCP Handshake Analysis

Observed SYN, SYN-ACK, and ACK packets during TCP session establishment.

4. HTTP Traffic Analysis

Hosted a Python HTTP server and analyzed HTTP GET requests and responses.

5. Reconnaissance Detection

Performed Nmap scans and identified filtered ports caused by firewall filtering.

6. Service Enumeration

Exposed a web service on TCP 8081 and used Nmap version detection to identify:

- HTTP Service
- SimpleHTTPServer 0.6
- Python 3.10.10

Key Findings

- ARP is required before IP communication.
- TCP uses a three-way handshake.
- Firewalls can cause ports to appear filtered.
- Nmap SYN scans generate SYN → SYN-ACK → RST patterns.
- Service enumeration reveals software and version information useful to attackers.

Skills Demonstrated

- Wireshark Packet Analysis
- TCP/IP Networking
- ARP Investigation
- ICMP Troubleshooting
- Nmap Reconnaissance
- Service Enumeration
- Security Documentation